

Coding & Architecture

System Design, Microservices, Security, Performance & Clean Code

SYSTEM ARCHITECTURE & SETUP

1. Scalable Project Boilerplate Generation

Act as a Senior Software Architect. Generate a production-ready, highly scalable directory structure and architecture guidelines for [Insert Tech Stack].

2. Microservices vs Monolith Decision Framework

Compare monolithic vs microservices tradeoffs covering deployment complexity, database sharing rules, and inter-service latency.

3. Serverless Backend Architecture Blueprint

Design a serverless backend addressing cold starts, database connection pooling, state persistence, and webhook reliability.

4. Authentication & RBAC Flow Design

Design an enterprise RBAC and token architecture with secure storage, refresh token rotation, and middleware authorization.

PERFORMANCE, SECURITY & APIS

5. Complex SQL Query Optimization

Rewrite slow queries using EXPLAIN ANALYZE, composite indexes, CTEs, and eliminating table scans.

6. Real-Time WebSockets Architecture

Design scalable real-time WebSocket connection pools with Redis pub/sub adapters, heartbeats, and reconnection handling.

7. OWASP Top 10 Security Hardening

Provide strict mitigation for SQLi, XSS, SSRF, and BOLA/IDOR vulnerabilities in enterprise APIs.

8. Distributed Caching & Invalidation

Implement cache-aside, write-through, probabilistic early expiration (XFetch), and CDC-based invalidation.